

CSE402: Optimisation

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1 Lecture 01

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Mathematical language of this course is linear algebra (of real vector spaces) and multivariable calculus. In this lecture, we will give a brief review of linear algebra.

§1.1 Review on linear algebra

Notation. Let $a_1, a_2, \dots, a_n$ be a finite sequence of real numbers. We will denote the sum

$$S = a_1 + a_2 + \dots + a_n \quad (1.1)$$

by

$$S = \sum_{k=1}^n a_k. \quad (1.2)$$

We have the following useful identities:

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad (1.3)$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}. \quad (1.4)$$

§1.1.1 Vectors

Definition 1.1.1 ($\mathbb{R}$ -vector space). Let V be a non-empty set. The set V equipped with two binary operations

$$+ : V \times V \rightarrow V$$

called *vector addition*, and

$$\cdot : \mathbb{R} \times V \rightarrow V$$

called *scalar multiplication* is called a **vector space over $\mathbb{R}$** or **$\mathbb{R}$ -vector space** if for all $u, v, w \in V$ and all $a, b \in \mathbb{F}$, the following axioms hold:

1. **Closure under addition:**

$$u + v \in V \quad (1.5)$$

2. **Commutativity of addition:**

$$u + v = v + u \quad (1.6)$$

3. **Associativity of addition:**

$$(u + v) + w = u + (v + w) \quad (1.7)$$

4. **Existence of zero vector:** There exists a vector $0 \in V$ such that

$$u + 0 = u \quad (1.8)$$

5. **Existence of additive inverse:** For every $u \in V$, there exists $-u \in V$ such that

$$u + (-u) = 0 \quad (1.9)$$

6. **Closure under scalar multiplication:**

$$au \in V \quad (1.10)$$

7. **Distributivity of scalar multiplication over vector addition:**

$$a(u + v) = au + av \quad (1.11)$$

8. **Distributivity of scalar multiplication over field addition:**

$$(a + b)u = au + bu \quad (1.12)$$

9. **Associativity of scalar multiplication:**

$$a(bu) = (ab)u \quad (1.13)$$

10. **Identity element of scalar multiplication:**

$$1u = u \quad (1.14)$$

Hence, $(V, +, \cdot)$ is called a vector space over the field $\mathbb{F}$. Elements of V are called **vectors** and elements of $\mathbb{R}$ are called **scalars**.

Example 1.1.2

Theorem 1.1.3

Blah

§1.1.2 Linear transformations